

Concurrency budget: where a 70B host's 640 GB pool goes

AGGREGATE RESIDENCY SCREEN $\neq$ PER-RANK FIT GUARANTEE

scenario reserve
(not a hardware constant)

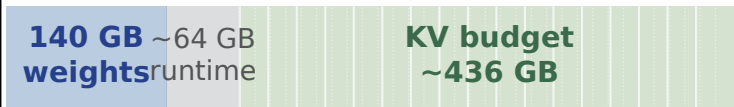
FP16 (2.62 MB/token, 9.5K max)

FP16 KV



$C \approx 18$ concurrent requests @ FP16

FP8 KV



$8 \times H100 = 640 \text{ GB}$

FP8 KV: ~ 33 slots ($436 \div 13.4 \text{ GB/request}$)

$8 \times 80 \text{ GB}$ per-rank pool (sharding/fragmentation/workspace still limit the real per-rank allocation; this is an aggregate screen)

0 100 200 300 400 500 600 700 800
on-host HBM (GB, per $8 \times H100$ host)